Code No: 111AL **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, June - 2022
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) From the data find number of students who obtained marks between 40 and 45.

Marks	30-40	40-50	50-60	60-70	70-80
Number of students	31	42	51	35	31

b) Using Lagrange's interpolation, find the interpolating polynomial for the data: [8+7]

X	0	1	3
F(X)	1	3	55

2.a) Using the least square method fit a straight line for the following data:

X	-1	0	1	2
Y(X)	1	0	1	4

b) Using the least square approximation fit a quadratic polynomial for the data: [8+7]

X	0	1	2
Y(X)	1	6	17

3.a) Using the Newton Raphson Method, find the cube root of 12.

b) Solve the following system by Gauss-Seidel method:

$$10x_1 - 2x_2 - x_3 - x_4 = 3$$
$$-2x_1 + 10x_2 - x_3 - x_4 = 15$$
$$-x_1 - x_2 + 10x_3 - 2x_4 = 27$$
$$-x_1 - x_2 - 2x_3 + 10x_4 = -9$$ [8+7]

4.a) The velocity V(km/min) of a moped which starts from rest is given at fixed intervals of the time T(min) as follows:

T	2	4	6	8	10	12	14	16	18	20
V	10	18	25	29	32	20	11	5	2	0

Using the 1/3 Simpson's rule estimate the distance covered in 20 minutes.

b) Using 4th order Runge Kutta Method, with h =0.1 find y(0.1) from $y' = y - x$ with $y(0) = 2$ [8+7]

5. Expand $f(x) = x\sin x, \quad 0 < x < 2\pi$ as a Fourier series. [15]

6. A tightly stretched string of length has its ends fastened at x=0 and x=L. The mid- point of the string is them taken to height h and then released from rest in that position. Find the expression for the displacement of the string at any point at any time. [15]

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7. A bar of 10 cms long with insulated sides has its ends A and B maintained at temperature 50 ^{0}C and 100 ^{0}C respectively until the steady state conditions prevail. The temperature at A is suddenly raised to 90 ^{0}C and at the same time that at B is reduced to 60 ^{0}C. Find the temperature distribution in the bar at any point at any time. [15]

8. Verify the Stroke's theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$, and $y = b$. [15]

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